SS1 Science Track: Physics Diagnostic (Part 1)

Curriculum: Cambridge IGCSE / O-Level Standard
Cluster 1: Measurements & Mechanics (Questions 1 - 20)

Constants: $g = 10 \text{ m/s}^2$ (unless otherwise stated). Use $\rho = \frac{m}{V}$, $v = \frac{d}{t}$, $F=ma$, $W=Fd$.

No.	Question & Options	Ans	Diagnostic Focus
1	Which instrument is most suitable for measuring the diameter of a thin copper wire? A) Metre rule B) Vernier caliper C) Micrometer screw gauge D) Tape measure	C	Precision and Instrumentation.
2	A block of metal has a mass of 540g and a volume of 200 cm^3. Calculate its density. A) 2.7 g/cm^3 B) 1.08 g/cm^3 C) 740 g/cm^3 D) 0.37 g/cm^3	A	Density & Mass-Volume ratio.
3	A car accelerates from rest to 20 m/s in 5 seconds. What is its acceleration? A) 4 m/s^2 B) 100 m/s^2 C) 5 m/s^2 D) 0.25 m/s^2	A	Kinematics (Acceleration formula).
4	In a speed-time graph, a horizontal straight line indicates: A) Constant speed B) Deceleration C) Constant acceleration D) The object is at rest	A	Graphical Analysis of Motion.
5	Calculate the weight of an object with a mass of 5 kg on Earth. A) 5 N B) 0.5 N C) 50 N D) 500 N	C	Distinction between Mass vs. Weight.
6	Which of these is a scalar quantity? A) Velocity B) Displacement C) Force D) Speed	D	Vectors vs. Scalars foundation.
7	A force of 15 N is applied to a spring, causing it to stretch by 3 cm. Find the spring constant (k). A) 5 N/cm B) 45 N/cm C) 0.2 N/cm D) 18 N/cm	A	Hooke's Law application.
8	Which of Newton's Laws states that for every action there is an equal and opposite reaction? A) First Law B) Second Law C) Third Law D) Law of Universal Gravitation	C	Newtonian Mechanics (Laws).
9	Calculate the work done when a force of 100 N moves an object 5 m in the direction of the force. A) 20 J B) 500 J C) 105 J D) 0.05 J	B	Work, Energy, and Power.
10	What is the power of a machine that does 1200 J of work in 1 minute? A) 1200 W B) 20 W C) 60 W D) 72000 W	B	Work/Time ratio (Time unit conversion check).

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11	Efficiency is defined as: A) Total Input / Useful Output B) Useful Output / Total Input C) Useful Output $\times$ Total Input D) Waste Energy / Work Done	B	Energy transformations & Efficiency.
12	When a mercury-in-glass thermometer is placed in hot water, the mercury level first drops slightly before rising. Why? A) Mercury contracts when first heated B) The glass bulb expands before heat reaches the mercury C) The water cools the glass initially D) Mercury is heavier than glass	B	Thermal Expansion (Conceptual).
13	A person of weight 600 N stands on one foot. If the area of the foot is 0.02 m^2, calculate the pressure exerted on the floor. A) 12 Pa B) $30,000 \text{ Pa}$ C) $12,000 \text{ Pa}$ D) 0.0003 Pa	B	Pressure calculations ($P=F/A$).
14	Which of these processes describes heat transfer through a vacuum? A) Conduction B) Convection C) Radiation D) Evaporation	C	Thermal Transfer mechanisms.
15	Find the kinetic energy of a 2 kg object moving at 10 m/s. A) 20 J B) 100 J C) 200 J D) 50 J	B	Kinetic Energy Formula ($\frac{1}{2} mv^2$).
16	A see-saw is balanced. A boy of 400 N sits 2m from the pivot. How far from the pivot must a girl of 200 N sit to balance him? A) 1m B) 2m C) 4m D) 0.5m	C	Moments and Equilibrium.
17	If a gas is compressed at a constant temperature, its pressure increases because: A) The molecules move faster B) The molecules hit the walls more frequently C) The molecules get larger D) The molecules stick together	B	Gas Laws (Kinetic Theory).
18	Vector addition: Two forces, 3N and 4N, act at right angles. The resultant force is: A) 7N B) 1N C) 5N D) 12N	C	Vector mathematics (Pythagoras).
19	Which unit is equivalent to a Joule per second (J/s)? A) Newton B) Pascal C) Watt D) Volt	C	Unit derivation & consistency.
20	**Diagnostic Insight:** Failure in Q3, Q10, and Q13 indicates the student lacks the mathematical dexterity for Cambridge Physics and should be monitored for the Science track.	-	Quantitative reasoning check.